

Blockchain Assignment 1 Narayan 32

```
import hashlib
```

```
def get_sha256(text: str) -> str:  
    # Hash functions in hashlib require byte inputs  
    return hashlib.sha256(text.encode("utf-8")).hexdigest()
```

```
# Question 1: User input and hash generation  
user_input_1 = input("Enter the original text: ")  
hash_1 = get_sha256(user_input_1)
```

```
print(f"\nOriginal Input: '{user_input_1}')
```

```
print(f"SHA-256 Hash 1: {hash_1}")
```

```
# Question 2: Modified input with one character changed  
user_input_2 = input("\nEnter modified text (change just one character): ")  
hash_2 = get_sha256(user_input_2)
```

```
print(f"\nModified Input: '{user_input_2}')
```

```
print(f"SHA-256 Hash 2: {hash_2}")
```

```
... Enter the original text: 12jj  
  
Original Input: '12jj'  
SHA-256 Hash 1: fe6da8a743048be55ea3cdabfafea9a6e00054e8a6454f0a7e06d295c940c  
  
Enter modified text (change just one character): 4545  
  
Modified Input: '4545'  
SHA-256 Hash 2: a994696540befd55c96017a162c7ae2685f2010a7fd3224c0ada252419139
```